PulsePatch Lite

Pulse Summary Report

Date: Oct 31, 2026
User: Volunteer VOL_BLE_SIM_01
Range: Oct 26, 2026 - Oct 31, 2026

EDUCATIONAL PROTOTYPE - NON-DIAGNOSTIC USE ONLY

PulsePatch Lite is an educational biomonitoring prototype connecting to an ESP32-C3 + MAX30102 pulse sensor. This report contains measured optical pulse rates, rule-based summary statistics, and experimental ML pattern labels. It does NOT constitute a medical diagnosis or health evaluation.

Aggregate Metrics

Metric	Value	Unit	Classification
Total Sessions	3	sessions	Measured
Eligible Analyzed Sessions	3	sessions	Calculated
Overall Mean Pulse Rate	74	BPM	Calculated
Resting Mean Pulse Rate	72	BPM	Calculated
Post-Activity Mean Pulse Rate	75	BPM	Calculated

Session Records Log

Timestamp	Protocol	Duration	Mean HR	SpO2	Quality	AI Pattern	Source
2026-10-26 09:30	Rest	60s	72 BPM	N/A	good	Typical pattern	Hardware
2026-10-26 10:43	Post-Activity	90s	74 BPM	N/A	good	Typical pattern	Hardware
2026-10-26 13:22	Post-Activity	90s	76 BPM	N/A	good	Different pattern	Hardware

Data Origin Separation:

- Received Data: Real over-the-air BLE packets from test peripheral (PulsePatch-SimHR) streaming simulated physiological profiles; physical MAX30102 sensor integration pending.
- Calculated: Rule-based median, mean, standard deviation, and interval arithmetic.
- Model Output: On-device machine learning pattern classification (Typical / Different). Strictly educational, non-diagnostic.

Authoritative Clinical Evidence Bases:

- WHO Guidelines on Hypertension (2023) & Pulse Oximetry (2021)
- ESH/ESC Guidelines for Arterial Hypertension (2023)
- AHA/ACC Scientific Consensus on Resting Heart Rate (2022)
- IEEE 1708 Standard for Cuffless Blood Pressure Measurement Boundaries